

GreenOps AI: Xebia’s Unified Carbon Reduction Dashboard

This proposal outlines the need and solution for the **GreenOps AI Dashboard**, a strategic initiative designed to transition Xebia from manual, reactive environmental tracking to a **proactive, data-driven sustainability model**. The core solution is a unified platform that integrates decentralized cloud consumption data (AWS/Azure) with operational data, uses **AI to forecast carbon emissions**, and provides engineers with actionable, "Shift-Left" recommendations to significantly reduce Xebia’s carbon footprint and optimize cloud spend.

The Challenge: Visibility and Velocity in ESG

Modern IT consulting firms face a critical challenge: a lack of transparent, real-time linkage between their primary operational engine (Cloud Infrastructure) and their sustainability commitments (ESG).

- 1. **Decentralized Footprint:** Xebia’s cloud consumption across multiple providers (AWS, Azure) and regions creates disparate data silos, preventing accurate, real-time measurement of the actual carbon cost of our services.
- 2. **Manual Reporting:** Current data aggregation for Environmental, Social, and Governance (ESG) reporting is often manual and retrospective, which is too slow to influence infrastructure decisions.
- 3. **Lack of Predictive Intelligence:** Without forecasting, Xebia’s teams cannot anticipate future carbon spikes or proactively identify resource-intensive applications, leading to reactive and inefficient optimization efforts.

Proposed Solution: GreenOps AI Dashboard

The GreenOps AI Dashboard is an MVP designed to serve as Xebia's central nervous system for sustainability, focusing on operationalizing environmental responsibility.

Key Deliverables:

Component	Function	Technical Focus
Data Aggregation Engine	Ingests and normalizes simulated usage data from AWS/Azure billing APIs, correlating cloud services (VMs, storage, networking) with estimated power consumption and CO2e.	Cloud Connectivity (AWS/Azure), Data Pipelines (Python/Serverless)
AI Forecasting Model	Implements a Time Series Machine Learning model (e.g., ARIMA or Regression) trained on historical usage patterns to predict Xebia's carbon footprint for the next quarter.	AI/ML Implementation, Predictive Analytics, Model Accuracy

Unified ESG Dashboard	Visualizes key, real-time metrics including Total Carbon Emissions (CO2e), Cloud Cost per Kg CO2e, and Carbon Intensity across business units or projects.	Front-end Development (React/Streamlit), Data Visualization
Optimization Engine	Provides data-driven, actionable recommendations (e.g., "Rightsizing non-prod VMs," "Switching to cheaper, lower-carbon regions") based on the AI forecast.	Business Logic, Recommendation Algorithm Design

Operationalizing Sustainability: The DevOps “Shift-Left” Strategy

A core innovation of the GreenOps AI Dashboard is the concept of a **"Shift-Left" Sustainability Score**.

The project will simulate integrating a **"Green Score"** (A-F) directly into a project's build process, much like **DevOps Shift-Left Security**.

- **The Scoring System:** The dashboard will automatically assign an environmental score to an application or service based on its predicted or measured carbon intensity.
 - **Pipeline Integration:** In a simulated CI/CD pipeline, a project with a low Green Score (D or F) would trigger an automatic warning or a soft gate.
 - **Goal:** This mechanism ensures that Green Coding and optimal resource utilization are addressed by developers *before* the code is deployed to production, thereby operationalizing sustainability and avoiding costly, carbon-heavy fixes later.
-

Expected Impact and Outcomes

This project delivers high-impact results directly supporting Xebia’s strategic goals:

- **Quantifiable Carbon Reduction:** Provides the necessary data to achieve a measurable reduction in Xebia’s operational carbon footprint.
- **Cost Efficiency:** By identifying resource-heavy applications, the tool inherently optimizes cloud consumption, converting a reduction in CO2e directly into savings on the monthly cloud bill.
- **Market Leadership:** Demonstrates Xebia’s technical expertise in GreenOps and Agentic DevOps, creating a powerful internal tool and a valuable, repeatable solution for clients who are also struggling with their cloud sustainability.
- **New Learning IP:** The project output can be immediately used as a case study and training module within Xebia Academy, reinforcing our commitment to relevant, cutting-edge curriculum.